Sahaj Bhadja

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Summary

Systems-oriented software engineer experienced in **Python, C/C++, distributed systems, Docker, SQL/NoSQL, testing, and performance optimization**, delivering reliable core services in collaborative, fast-moving engineering environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built distributed Python services processing 5,000+ monthly jobs with explicit state transitions, failure isolation, validation, and reconciliation, increasing concurrent throughput 3x while reliably sustaining 99%+ production reliability.
- Optimized PostgreSQL and SQL Server schemas, indexing, caching, and access patterns for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds and integration latency 35%.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Implemented multithreaded Python services and REST APIs processing 1,000+ files per batch, adding fault-tolerant execution, diagnostics, fallback paths, and automated tests for stable peak-load performance at scale.
- Investigated production issues across 10,000+ daily requests, resolving OCR, model, and API failures while collaborating on modular Flask components, code quality, and reliable feature delivery at scale.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Developed performance-sensitive Python and PyTorch components processing 100,000+ trajectory records, improving long-horizon planning 18% through rapid prototyping, simulation, debugging, reproducible testing, and iterative system optimization at scale.
- Collaborated with senior researchers to analyze edge cases and system interactions, increasing closed-loop stability 15% while documenting implementation tradeoffs, experimental findings, and reproducible technical decisions at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Java and Python microservices using Docker, SQL, AWS Glue, and S3, improving API response times 30%, reducing payload sizes 25%, and lowering storage costs 15%.
- Owned requirements-through-deployment improvements, automating 20+ tests across Windows, macOS, and Linux while increasing pipeline efficiency 68% and collaborating with Microsoft and GitHub stakeholders during Agile peer reviews.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built eight distributed Node.js services for authentication, resource allocation, customer records, billing, permissions, and notifications, reliably synchronizing two million records through WebSockets, REST APIs, and Azure SQL.
- Automated testing and Docker-based deployments through Jenkins, reducing release time 50% while using code reviews, monitoring, and operational analytics to improve reliable system delivery across release cycles.

Skills

Languages: Python, C, C++, Java, SQL, Shell, Bash, TypeScript, JavaScript
Systems: Distributed systems, operating systems, concurrency, multithreading, fault tolerance, caching
Data/Cloud: PostgreSQL, MySQL, MongoDB, NoSQL, Docker, Kubernetes, AWS, Azure, S3, microservices
Quality: Git, CI/CD, Jenkins, PyTest, Mockito, unit testing, integration testing, debugging, peer review

Projects

CPU-scheduling Simulator Live | *C++, Systems Software*
- Implemented six C++ scheduling algorithms with edge-case handling, automated waiting and turnaround metrics, and Gantt visualization, testing 1,000+ process combinations while reliably reducing average turnaround time 25%.

Chat2SQL Code | *Docker, SQL/NoSQL, Cross-Platform Systems*
- Developed Java and TypeScript services supporting 10+ SQL and NoSQL databases, configuring Docker workflows across Windows, macOS, and Linux with schema synchronization, validation, and reliable local debugging.

FinNexus Live | *Distributed Platform, Containers*
- Built a distributed fintech platform supporting 50,000+ users and one million transactions using TypeScript, MongoDB, WebSockets, Kubernetes, secure token controls, real-time synchronization, and reliable automated deployment workflows.